

FAISAL AL-DROU

AI & ML ENGINEER

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PROFESSIONAL SUMMARY

Artificial Intelligence and Robotics Engineer with proven experience in building end-to-end Generative AI applications, Retrieval-Augmented Generation (RAG) pipelines, and deep learning solutions. Skilled in LLM orchestration (LangChain, LCEL), autonomous agent development, and production API deployment, alongside applied experience in machine learning and robotics systems. Ranked 2nd academically, with a strong ability to architect scalable, data-driven AI systems that automate operations, enhance customer conversion, and drive business growth.

TECHNICAL SKILLS

Artificial Intelligence & Generative AI:	Machine Learning, Deep Learning, Neural Networks, LangChain Expression Language (LCEL), Multi-Provider Model Routing, Prompt Engineering, Predictive Modeling, Retrieval-Augmented Generation (RAG), Autonomous Conversational Agents
AI Frameworks & Libraries:	LangChain, Pydantic v2, Groq, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, CatBoost, Optuna
Data Science & Output Parsing:	Pandas, NumPy, JSON Output Parsing, Feature Engineering, Data Preprocessing, Data Transformation, Data Analysis, Apache Parquet
Web & Middleware Architecture:	Python (Flask), RESTful API Design, Middleware Architecture, Error Handling & Fallback Pipelines, Latency Tracking & Benchmarking, Twilio API (Webhooks), Gunicorn
Programming Languages:	Python, SQL, C++, JavaScript (Fetch API / Async), Kotlin, Arduino C
Development Tools & Security:	Git, Linux, Bash, Ubuntu Command Line, Environment Variable Management (python-dotenv), Secret Isolation, GitHub, Render (Cloud Hosting)

PROFESSIONAL EXPERIENCE

Al-Namothaj School for Special Education — *Computer Science & Robotics Instructor* Amman, Jordan
Delivered practical training in computer science and robotics, enabling students to understand programming concepts through hands-on activities.

- Developed simplified learning materials for complex technical concepts, improving students' understanding and application of programming skills.
- Designed interactive technical activities that enhanced students' problem-solving abilities and computational thinking skills.

Infotel — *Machine Learning Intern* 2023
Participated in hands-on practical training focusing on data science workflows, exploratory data analysis, and baseline machine learning implementations.

- Conducted exploratory data analysis (EDA) and feature preprocessing on structured datasets using Python, Pandas, and NumPy to support modeling readiness.
- Assisted in running baseline model experiments using Scikit-Learn to compare standard classification metrics across test datasets.

TECHNICAL PROJECTS

Autonomous GenAI Sales & WhatsApp RAG Web Application

Technologies: Python (Flask), LangChain (LCEL), Twilio API, Pydantic v2, REST API, Groq, Meta Llama, Google Gemini, JavaScript, Gunicorn

Architected an end-to-end RAG assistant supporting multi-turn conversational memory, localized dialect processing (Jordanian Arabic), and automated sales pipeline progression.

- Integrated bi-directional WhatsApp Webhooks via Twilio API, enabling real-time, asynchronous mobile customer communication with sub-second AI response delivery.
- Engineered a high-performance knowledge retrieval pipeline using header-aware document chunking and LangChain retrievers to dynamically ground LLM responses in verified pricing and operational rules.
- Implemented deterministic lead extraction using Pydantic v2 schemas and JsonOutputParser to autonomously detect purchase intent, capture client metadata (name, phone, tier), and persist qualified leads.
- Enforced strict closed-domain guardrails within LangChain LCEL pipelines to eliminate hallucinations, prevent unauthorized external URLs, and gracefully handle out-of-scope inquiries.
- Deployed the production microservice on Render/Gunicorn with multi-provider LLM fallbacks (Groq/Llama-3.1, GPT-OSS, Gemini) to eliminate rate-limit degradation and ensure 24/7 availability.

ClimaPredict AI – Sub-Seasonal Temperature Forecasting System

Technologies: Python, LightGBM, XGBoost, CatBoost, Scikit-Learn, Streamlit, Plotly, Optuna, Pandas, NumPy, Joblib

- Developed and tuned a LightGBM sub-seasonal temperature forecaster across 240+ meteorological features, reaching a 99.65% R-squared score (0.5738 RMSE) compared to baseline linear models.
- Built and deployed a production-ready Streamlit web app featuring dynamic batch CSV schema alignment, real-time single inference via custom imputation, and interactive Plotly exploratory dashboards.

New York City Collision Data ETL Pipeline

Technologies: Python, Pandas, NumPy, JSON, Apache Parquet

Built a data processing pipeline to clean, transform, and organize large-scale collision datasets, improving data quality and preparing it for analysis and insight extraction.

Binary Image Classification Using Convolutional Neural Networks (CNN)

Technologies: Python, TensorFlow, Keras, NumPy, Matplotlib

Developed a deep learning model for automated image classification, enabling pattern recognition through convolutional neural network techniques.

EDUCATION

Artificial Intelligence and Robotics Engineering

GPA: 83.3% | Academic Rank: Ranked 2nd in class

Graduation Date: 2024

TECHNICAL TRAINING & CERTIFICATIONS

IBM RAG and Agentic AI Professional Certificate (In Progress) – Coursera / IBM

- Completed: "Develop Generative AI Applications" (LangChain, LCEL, Pydantic, Flask)
- Completed: "Build RAG Applications" (LlamaIndex, RAG, Embeddings, Gradio, Document Management)

Data Science and Artificial Intelligence Training – SHAI for AI

Artificial Intelligence and Machine Learning in Medical Applications – Course The Hope International

Python for Machine Learning Certificate – Pro Vision

LANGUAGES

Arabic: Native • English: Advanced • Japanese: Intermediate